

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12387974
B2
August 12, 2025
Lo; Wei-Jen et al.

Patterning interconnects and other structures by photo-sensitizing method

Abstract

A representative method includes forming a photo-sensitive material over a substrate, and forming a cap layer over the photo-sensitive material, and patterning the cap layer. Using the patterned cap layer, a first portion of the photo-sensitive material is selectively exposed to a pre-selected light wavelength to change at least one material property of the first portion of the photo-sensitive material, while preventing a second portion of the photo-sensitive material from being exposed to the pre-selected light wavelength. One, but not both of the following steps is then conducted: removing the first portion of the photo-sensitive material and forming in its place a conductive element at least partially surrounded by the second portion of the photo-sensitive material, or removing the second portion of the photo-sensitive material and forming from the first portion of the photo-sensitive material a conductive element electrically connecting two or more portions of a circuit.

Inventors: Lo; Wei-Jen (Hsinchu, TW), Shih; Po-Cheng (Hsinchu, TW), Jang; Syun-Ming (Hsinchu, TW), Lee; Tze-Liang (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Family ID: 76882863

Assignee: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Appl. No.: 18/738256

Filed: June 10, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240332068 A1	Oct. 03, 2024

Related U.S. Application Data

continuation parent-doc US 18309131 20230428 US 12033890 child-doc US 18738256
continuation parent-doc US 17094700 20201110 US 11676855 20230613 child-doc US 18309131
us-provisional-application US 62981862 20200226

Publication Classification

Int. Cl.: **H01L21/768** (20060101); **G03F7/038** (20060101); **G03F7/039** (20060101); **G03F7/20** (20060101); **H01L21/027** (20060101)

U.S. Cl.:

CPC **H01L21/76823** (20130101); **G03F7/038** (20130101); **G03F7/039** (20130101); **G03F7/2004** (20130101); **G03F7/2022** (20130101); **H01L21/0274** (20130101); **H01L21/76802** (20130101); **H01L21/76877** (20130101);

Field of Classification Search

CPC: H01L (21/76823); H01L (21/0274); H01L (21/76802); H01L (21/76877); H01L (21/76885); H01L (21/76832); H01L (21/76834); H01L (21/76837); H01L (21/76849); H01L (21/31058); H01L (21/311); H01L (21/31144); H01L (21/76816); H01L (2221/101); G03F (7/038); G03F (7/039); G03F (7/2004); G03F (7/2022)

USPC: 430/319; 438/597

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4608129	12/1985	Tamamura et al.	N/A	N/A
9235116	12/2015	Inasaki et al.	N/A	N/A
10861710	12/2019	Kuo et al.	N/A	N/A
11417604	12/2021	Yu et al.	N/A	N/A
11676855	12/2022	Lo et al.	N/A	N/A
12033890	12/2023	Lo	N/A	H01L 21/76823
2003/0203321	12/2002	Ma et al.	N/A	N/A
2004/0089878	12/2003	Takehashi et al.	N/A	N/A
2011/0065050	12/2010	Li et al.	N/A	N/A
2019/0198786	12/2018	Shimizu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1439117	12/2002	CN	N/A
107689333	12/2017	CN	N/A
102018125161	12/2018	DE	N/A
2004061629	12/2003	JP	N/A
2004087976	12/2003	JP	N/A
2005003803	12/2004	JP	N/A
2008227465	12/2007	JP	N/A
2013020226	12/2012	JP	N/A

900003155	12/1989	KR	N/A
432707	12/2000	TW	N/A
0206898	12/2001	WO	N/A

Primary Examiner: Sullivan; Caleen O

Attorney, Agent or Firm: Slater Matsil, LLP

Background/Summary

CLAIM FOR PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 18/309,131, filed on Apr. 28, 2023, entitled “Patterning Interconnects and Other Structures by Photo-Sensitizing Method,” which is a continuation of U.S. patent application Ser. No. 17/094,700, filed on Nov. 10, 2020, now U.S. Pat. No. 11,676,855 issued on Jun. 13, 2023, entitled “Patterning Interconnects and Other Structures by Photo-Sensitizing Method,” which claims priority to U.S. Provisional Patent Application No. 62/981,862, filed on Feb. 26, 2020, and entitled “Patterning Interconnects and Other Structures by Photo-Sensitizing Method,” which applications are incorporated herein by reference.

BACKGROUND

(1) Numerous steps in semiconductor and integrated circuit fabrication require patterning of layers, typically using photolithography techniques. As features become small and pattern density becomes correspondingly greater, new techniques are required to overcome drawbacks such as dry etch loading, complicated multi-layer mask requirements, inadvertent damage to the layer to be patterned or surrounding layers, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1a through 1f illustrate steps in forming a patterned photo-sensitive layer and the structures resulting therefrom.

(3) FIGS. 2a through 2d illustrate steps in forming a conductive feature in the patterned photo-sensitive layer of FIG. 1e, and the structures resulting therefrom.

(4) FIGS. 3a through 3f illustrate steps in forming a patterned photo-sensitive layer having conductive regions and non-conductive regions, and the structures resulting therefrom.

(5) FIGS. 4a through 4f illustrate steps in forming insulating features in the patterned photo-sensitive layer of FIG. 3f, and the structures resulting therefrom.

(6) FIGS. 5a through 5j illustrate steps in forming an electrical interconnect from a patterned photo-sensitive layer, and the structures resulting therefrom.

(7) FIGS. 6 and 7 illustrate properties of illustrative photo-sensitive materials.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or

on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) In the following described embodiments, a method of forming an interconnect feature, such as a conductive via, within a dielectric layer, such as an inter metal dielectric (IMD) is disclosed. The present teaching is equally applicable, however, to conductive lines within IMDs, conductive via and/or contact plugs within inter layer dielectric (ILD) layers, conductive pads, and the like.

(11) A first embodiment begins with a structure **10**, which is illustrated in FIG. **1a** at an intermediate stage of manufacture wherein one or more features and/or layers have been formed in and/or on a substrate **2**. Substrate **2** is shown in cross sectional view and, like the other features illustrated herein, is not drawn to scale unless specifically described as such. Generally, substrate **2** may comprise a bulk semiconductor substrate or a silicon-on-insulator (SOI) substrate. An SOI substrate includes an insulator layer below a thin semiconductor layer that is the active layer of the SOI substrate. The semiconductor of the active layer and the bulk semiconductor generally comprise the crystalline semiconductor material silicon, but may include one or more other semiconductor materials such as germanium, silicon-germanium alloys, compound semiconductors (e.g., GaAs, AlAs, InAs, GaN, AlN, and the like), or their alloys (e.g., $Ga_{x}Al_{1-x}As$, $Ga_{x}Al_{1-x}N$, $In_{x}Ga_{1-x}As$ and the like), oxide semiconductors (e.g., ZnO, SnO₂, TiO₂, Ga₂O₃, and the like) or combinations thereof. The semiconductor materials may be doped or undoped. Other substrates that may be used include multi-layered substrates, gradient substrates, or hybrid orientation substrates. Only a small portion of substrate **2** (as well as subsequently formed layers) is illustrated in the accompanying figures for the sake of simplicity and clarity, as such disclosure is sufficient for understanding the embodiments described herein.

(12) One or more active and/or passive components (not illustrated) may also have been formed prior to the state illustrated in FIG. **1a**. Components such as FinFET transistors, planar transistors, Gate All Around (GAA) transistors, capacitors, resistors and the like are within the contemplated scope of devices included within structure **10**. In addition, local interconnects formed of conductive polysilicon or other conductive materials, inter layer dielectrics (ILDs), contact plugs, lower levels of interconnects such as metal layers embedded within respective dielectric layers, and the like are all within the contemplated scope of substrate **2** represented in FIG. **1a**.

(13) In this particular embodiment, a dual etch stop feature, including etch stop layer (ESL) **4** and ESL **6**, is shown having been formed over substrate **2**. The distinction between substrate **2** and layers formed over substrate **2** is somewhat arbitrary and for illustration only. In other words, in some embodiments, ESL **4** and/or ESL **6** could be considered as part of substrate **2**, whereas in FIG. **1a** they are illustrated separately. Those skilled in the art will appreciate that ESLs such as ESL **4** and ESL **6** are employed to protect underlying features/layers/structures of substrate **2** from subsequent processes, particularly subsequent etching and/or patterning processes, as will be demonstrated in the following paragraphs. In the present example, two ESLs, ESL **4** and ESL **6**, are employed to ensure sufficient protection for the underlying features/layers/structures. This may be

necessitated because of the size, complexity, shape, pattern density, material compositions, and the like of those underlying features/layers/structures and/or the size, complexity, shape, pattern density, material compositions, and the like of a subsequently formed material (such as a subsequently formed IMD layer described in the following paragraphs), and the etch/patterning processes used to pattern the subsequently formed material. In other embodiments, a single ESL may be sufficient and in yet other embodiments three or more ESLs might be required for sufficient protection of underlying features/layers/structures.

(14) As but one example, ESL **4** may be an aluminum oxide (Al_xO_y) layer and ESL **6** may be an oxygen doped silicon carbide (sometimes referred to as oxygen doped carbide or “ODC”) layer. While ODC layer **6** is illustrated as being formed over Al_xO_y layer **4**, it is within the contemplated scope of this application that Al_xO_y layer **4** could alternatively be formed over ODC layer **6**.

(15) Continuing with FIG. **1b**, a layer **8** is blanket deposited over ESL **4** and ESL **6**, illustratively being formed directly on ESL **6**. In an illustrative embodiment, dielectric layer **8** may serve as an IMD layer for the structure **10**, but this function is simply one of many applications for the concepts disclosed herein. In this embodiment, layer **8** comprises thiophene, phenyl, or thiophenyl, as examples. As will be discussed in greater detail below, layer **8** changes at least one material property when exposed to a sufficient dosage of light within a specified wavelength, and may be referred to herein as photo-sensitive layer **8**.

(16) Photo-sensitive layer **8** may be deposited using a Chemical Vapor Deposition (CVD) process, such as flowing a precursor gas such as one or more of ethane, an alkene, an alkyne, or the like, at a flow rate ranging from about 10 sccm to about 1000 sccm, at a temperature typically of from about 50° C. to about 300° C., for example. Alternatively, photo-sensitive layer **8** could be deposited by a different deposition process, such as PECVD, MOCVD, CVD, ALD, or the like. The photosensitive layer **8** could comprise a material such as I-R, P-R, or the like, where I is Iodonium salt, P is Phosphorous, and R represent C_xH_y , or a COOH organic group. The use of an O_2 or Cox plasma pretreat process will improve interface adhesion for the photosensitive layer **8** and hence will reduce or preferably eliminate peeling, particularly when the layer is deposited using CVD deposition processes.

(17) While the thickness of photo-sensitive layer **8** will depend upon the specific application and structure being manufactured, as an example photo-sensitive layer **8** may be deposited to a nominal 10 nm to about 60 nm thickness.

(18) FIGS. **1c** through **1f** illustrate steps for patterning photo-sensitive layer **8** by taking advantage of its photo-sensitive properties. First, as shown in FIG. **1c**, one or more cap layers are formed on the top surface of photo-sensitive layer **8**. In an illustrative embodiment, a first dielectric cap layer **12** is formed by tetraethyl orthosilicate deposition (sometimes referred to as tetraethoxysilane, and commonly abbreviated as TEOS), followed by a second cap layer **14**, formed of a metal such as tungsten, tungsten doped carbide, TiN , or the like. A third cap layer **16**, such as another TEOS dielectric layer is then formed, followed by a fourth cap layer **18**, such as an amorphous silicon dielectric cap layer. The specific materials and deposition processes disclosed for cap layers **12**, **14**, **16**, and **18** are by way of example only. The use and composition of the cap layers is a matter of design choice and depends upon factors such as etch selectivity of the underlying materials and the cap layer materials to the selected etch process, the required resolution and density of the resulting pattern, and other such process parameters. Various other materials and deposition processes are within the contemplated scope of this disclosure, including a lesser or a greater number of cap layers, provided that the combination and number of cap layers are sufficient to provide for the patterning function described with respect for FIGS. **1d** through **1e**. For instance, other materials such as SiO_2 , SiOC , SiCO , SiN , and the like could readily be used for one or more of the cap layers, which would be deposited to a thickness of perhaps about 5 nm to about 70 nm in a typical embodiment.

(19) Continuing the discussion with FIG. **1d**, illustrated therein is the result of a first patterning step

wherein cap layers **12** and **14** are patterned to form an opening or trench **19** extending therethrough. This can be accomplished, e.g., by forming and patterning a photoresist layer (not shown) using well-known lithographic techniques to etch a pattern into cap layers **16** and **18** and then in a same or separate etch process etch the pattern into cap layers **12** and **14** to obtain the structure illustrated in FIG. **1d**. Etch processes and compositions for etching through the demonstrative cap layers are well known and the details of which are not necessary to understanding the inventive concepts disclosed herein. Hence, those details are omitted for the sake of brevity and clarity. In the illustrated embodiment, cap layers **16** and **18** are removed (etched away) by the process of patterning cap layers **12** and **14**. In other embodiments portions of cap layers **16** and **18** might remain, or even the entire cap layers might remain and be patterned similarly to cap layers **12** and **14**. In still other embodiments, different patterning techniques, such as e-beam lithography, ion milling, and the like could be employed to pattern one or more of cap layers, **12**, **14**, **16**, and/or **18**. (20) As shown in FIG. **1d**, after cap layers **12** and **14** are patterned a region of photo-sensitive layer **8** is exposed through opening **19**, while remaining regions of photo-sensitive layer **8** remain unexposed, or covered, by cap layers **12** and **14**. Although only a single opening **19** is illustrated in FIG. **1d**, it should be apparent that numerous different openings of various sized and shapes can be simultaneously formed to expose numerous different regions of photo-sensitive layer **8**.

(21) A light exposure step is now performed, as schematically represented by arrows **20** in FIG. **1e**. Those regions of photo-sensitive layer **8** exposed through opening **19** are subjected to (exposed to) incoming light **20**, whereas cap layers **12** and/or **14** serve to block, absorb, and/or reflect incoming light **20** to keep the covered regions of photo-sensitive layer **8** from being subjected to (exposed to) incoming light **20**. Hence, photo-sensitive layer **8** is exposed in a pattern that corresponds to the pattern imposed upon cap layers **12** and **14**.

(22) In response to exposure to incoming light **20**, the exposed portions of photo-sensitive layer **8** change undergo a change in at least one material property. For instance, in the presently described embodiment, exposures to incoming light **20** causes an exposed region **22** of photo-sensitive layer **8** to undergo a change in its resistance to certain etch processes, such as by becoming significantly more susceptible to attack (removal) by the etch process. For instance, in the example wherein photo-sensitive layer **8** comprises thiophene, phenyl, thiophenyl, or the like, and incoming light **20** (which could be a light source, a laser, or the like) is in a wavelength range of about 200 nm to about 400 nm and at an exposure energy of perhaps about 10 mJ/cm² to about 100 mJ/cm², the exposed portions of photo-sensitive layer **8** become more susceptible to being etched by an etch process. While the specific etch selectivity between the exposed portion **22** and the unexposed portions of photo-sensitive layer **8** may vary and depend upon the particular application, an etch selectivity of from about 2× to about 50× is desirable.

(23) FIG. **1f** illustrates the results of one or more etch processes wherein exposed portions **22** of photo-sensitive layer **8** are removed. In one example, a first etch process, such as a wet etch using an organic acid or solvent is employed to remove exposed portions. This etch process will not significantly attack/remove ESL **6** (and/or ESL **4**) and hence the process “stops” on the ESLs. Next, a second etch process, such as using CF_x plasma, BCl₃ plasma, or the like is employed to remove the ESLs. The result is an opening **23** through which underlying layers/features/structures of substrate **2** are exposed for subsequent electrical and/or physical connection. Significantly, the unexposed portions of photo-sensitive layer **8** (i.e., the portions that were not exposed to light **20** and hence did not undergo a material change) are relatively un-etched by the etch processes that removed the exposed portions **22**. In other words, even though an anisotropic wet etch process was used to form opening **23**, because of the photo-exposure step described with reference to FIG. **1e**, the un-exposed portions of photo-sensitive layer **8** remain relatively un-etched and opening **23** will have relatively straight and vertical sidewalls. This allows for, e.g., greater packing density, tighter tolerances/process windows, and greater reliability of the resulting device.

(24) Various subsequent processing steps can next be employed to continue/complete manufacture

of device **10**. FIGS. **2a** through **2d** illustrate steps in one embodiment wherein a conductive via **26** is formed in opening **23**. For simplicity, substrate **2** is not illustrated in FIGS. **2a** through **2d**, although it would be so in actual product and processes. Beginning with FIG. **2a**, this figure illustrates the structure of FIG. **1f** after opening **23** has been over-filled with conductive material **24**. Conductive material **24** schematically represents one or more different layers and/or materials filling opening **23**. For instance, conductive material **24** could include one more liners lining sidewalls of opening **23** (e.g., Ti, Tin, Ta, TaN, and the like), as well as one or more fill materials such as cobalt, copper, aluminum, tungsten, gold, platinum, doped polysilicon, and the like. It is within the contemplated scope of this disclosure that even dielectric liners could fall within the rubric of conductive material **24**. In FIG. **2a**, conductive material extends to or just above the top of ESL **6**. In other embodiments (not specifically illustrated) conductive material **24** could extend over the top surfaces of ELS **6**. In yet other embodiments, ESL **6** and/or ESL **4** could be fully or partially removed before depositing conductive material **24**, in which case conductive material would overfill opening **23** and be at or above the top of photo-sensitive layer and possibly extending laterally on the top surface of photo-sensitive layer **8**.

(25) As illustrated in FIG. **2b**, any overfilled portions of conductive material **24** are removed, resulting in conductive feature **26** being formed in hole **23**. Conductive feature could just be, for instance, a conductive via, but could just as well be a contact, a plug, a wire, an interconnect, a pad or any other conductive feature. In most embodiments, conductive feature **26** electrically contacts an underlying feature, such as an underlying wire or interconnect, via, plug, or other feature such as a terminal for a transistor, capacitor, resistor, or the like. A planarization process such as a Chemical Mechanical Polish (CMP), an etch back, or the like can be employed to remove overfilled portions. Note that it is within the contemplated scope of the present disclosure that in some embodiments conductive material just fills, but does not overfill, opening **23**, in which case no planarization step such as illustrated in FIG. **2b** is needed.

(26) Proceeding to FIG. **2c**, in some embodiments an optional metal capping layer such as a cobalt cap is formed over conductive feature **26**. This metal capping layer may be particularly beneficial when conductive feature **26** is made of copper or in large part made of copper—in order to protect conductive feature **26** during subsequent processing steps. Optionally, additional conductive features can be manufactured by conventional techniques, or by repeating the processes illustrated in FIGS. **1a-1f** and **2a-2c**. For instance, as shown in FIG. **2d**, additional etch stop layers such as **4'** and **6'** can be formed overlying photo-sensitive layer **8** and conductive feature **26** and the above-described process repeated to form a subsequent conductive feature in a next level of interconnection (not shown) above the level shown in FIG. **2d**.

(27) FIGS. **3a** through **3g** illustrate another embodiment for forming conductive features in a photo-sensitive layer. Beginning with FIG. **3a**, a substrate **32** is provided, which substrate **32** can be the same or similar to substrate **2** described above. In other words, substrate **32** can be a bulk semiconductor wafer, a multi-layer wafer such as an SOI substrate, could comprise silicon or other semiconductor materials, could be doped or undoped, etc. Further, as described above with reference to substrate **2**, substrate **32** is contemplated encompass various layers/features/structures as may be formed in and on the wafer including but not limited to transistors, capacitors, resistors, interconnects, insulating layers, dielectric layer, metallization layers, and the like.

(28) ESL layers **34** and **36** are deposited atop substrate **32**. These ESL layers could be the same or similar to ESL layer **4** and **6** discussed above, but such similarity is not a requirement for this embodiment. Additionally, a single ESL layer or more than two ESL layers could be employed, provided the ESL layer or layers offer sufficient protection to underlying layers/features/structures during subsequent processing steps as described in the following paragraphs. FIG. **3b** illustrates a photo-sensitive layer **38** is deposited on ESL layer **6**. In this embodiment, photo-sensitive layer **38** is such that its conductivity changes significantly upon sufficient exposure to light energy of certain wavelengths. As an example, photo-sensitive layer **38** may be formed of phenyl or thiophenyl or

combinations of phenyl and thiophenyl, as examples. Such material is sensitive to and changes its conductivity when exposed to light energy in the range of about 13.5 nm (Extreme Ultra Violet or EUV range).

(29) In one example, photo-sensitive layer **38** is deposited using a CVD process such as flowing precursor gas such ethane, alkene, alkyne, or the like at a flow rate ranging from about 10 sccm to about 1000 sccm, and at a temperature typically of from about 50° C. to about 300° C., in an illustrative embodiment. Alternatively, photo-sensitive layer **38** could be deposited by a different deposition process, such as PECVD, MOCVD, CVD, ALD, and the like.

(30) In the embodiment illustrated in FIG. 3c, multiple cap layers are blanket deposited atop photo-sensitive layer **38**. In this case, first cap layer **42** and second cap layer **44** are made of material selected such that cap layers **42** and **44** are transparent or at least substantially transparent to light energy of the wavelengths to which photo-sensitive layer **38** is sensitive. For instance, in the case of phenyl and/or thiophenyl which are sensitive to light energy in the range described above, cap layers **42** and **44** are made from a material that is substantially transparent to that light wavelengths. For instance, cap layers **42** and/or **44** could be made of SiO₂, SiOC, or the like. The cap layers should be deposited to a sufficient thickness that that function to protect underlying photo-sensitive layer **38** during subsequent process steps (described below) but thin enough to allow all or substantially all light energy (of the appropriate wavelengths) to pass through during the exposure step described below with regard to FIG. 3f. For instance, cap layers **42** and **44** could be deposited to a thickness of from about 5 nm to about 70 nm, in an illustrative embodiment. Although two transparent cap layers are illustrated, a single transparent or more than two transparent cap layers are within the contemplated scope of this disclosure provided the above function constraints regarding transparency, etc. are met.

(31) FIG. 3c also illustrates two additional cap layers **46** and **48**. These cap layers are employed for patterning transparent cap layers **42** and **44** and generally do not remain after the transparent cap layers have been patterned—and hence there is not a need for these cap layers to be transparent. Hence, a wider range of materials, both conductive and dielectric could be used for cap layers **46** and **48**. In an embodiment, cap layer **46** is a TEOS oxide and cap layer **48** is amorphous silicon, although these layers could be used in reverse order.

(32) FIG. 3d illustrates the results of patterning transparent cap layers **42** and **44** to form opening **49** extending through transparent cap layer **42** and **44**. This can be accomplished, e.g., by forming and patterning a photoresist layer (not shown) using well-known lithographic techniques to etch a pattern into transparent cap layers **46** and **48** and then in a same or separate etch process etching the pattern into transparent cap layers **42** and **44** to obtain the structure illustrated in FIG. 3d. Etch processes and compositions for etching through the demonstrative cap layers are well known and the details of which are not necessary to understanding the inventive concepts disclosed herein. Hence, those details are omitted for the sake of brevity and clarity. In the illustrated embodiment, transparent cap layers **46** and **48** are removed (etched away) by the process of patterning transparent cap layers **42** and **44**. In other embodiments portions of cap layers **46** and **48** might remain, or even the entire cap layers might remain and be patterned similarly to transparent cap layers **42** and **44**. In still other embodiments, different patterning techniques, such as e-beam lithography, ion milling, and the like could be employed to pattern one or more of cap layers, **42**, **44**, **46**, and/or **48**.

(33) FIG. 3e illustrates the results of filling opening **49** with a plug **50**. Plug **50** should not be transparent to the light wavelengths to which photo-sensitive layer **38** is sensitive. A broad range of materials can hence be used for plug **50**, which will subsequently be removed. In an embodiment, plug **50** could be made from an oxide, such as silicon oxide for instance. Although only a single plug **50** is illustrated in FIG. 3d, it should be apparent that numerous different openings **49** of various sized and shapes can be formed and filled with numerous plugs **50** to cover different regions of photo-sensitive layer **38**. In some contemplated embodiments, plug **50** is formed through

a selective deposition process where the material comprising plug **50** forms readily on the exposed surface of photo-sensitive layer **38** and does not form readily on the exposed surface of cap layer **46**. In other contemplated embodiments, the material comprising plug **50** could be blanket deposited over the entire device and then patterned through a photolithographic process, a CMP or other planarization process, or the like. Contrary to the embodiment described above with regard to FIGS. **1a** through **1f**, opening(s) **49** is aligned with regions of photo-sensitive layer **38** that will remain after subsequent processing (whereas in the above embodiment, opening(s) **19** was aligned to regions of photo-sensitive layer **8** that were subsequently removed in order to form conductive feature(s) **26**. This is explained in the following paragraphs.

(34) After plug **50** is formed, a light exposure step is performed, as schematically represented by arrows **52** in FIG. **3f**. Light energy **52** is in pre-selected wavelength spectrum, such as EUV for instance. In other words, light energy **52** is of the wavelength that causes photo-sensitive layer **38** to change its conductivity properties when exposed thereto. The region of photo-sensitive layer **38** covered by plug **50** is protected from (not exposed by) light energy **52**. By contrast, the regions of photo-sensitive layer **38** covered by transparent cap layers **42** and **44** are exposed to light energy **52** because transparent cap layers **42** and **44** are substantially transparent to the wavelengths of light energy **52**. In response to exposure to incoming light **52**, the exposed portions of photo-sensitive layer **38** undergo a change in at least one material property and in this embodiment those exposed portions convert from a dielectric to a conductor **54**.

(35) Exposed regions **54** (now conductive) can be in the form of contact plugs, pads, elongated wires, or other forms of interconnect and the remaining unexposed regions **38** can be in the form of a patterned insulating layer that electrically isolated the various exposed conductive regions **54** from one another (for clarity, those portions of photo-sensitive layer **38** that remain unexposed will henceforth be called unexposed portions **38**, remaining portions **38**, or the like in contrast to those portions of photo-sensitive layer **38** that were exposed which will henceforth be called exposed regions **54**, conductive regions **54**, or the like). If the remaining unexposed regions **38** are sufficiently insulating to perform this function during operation of the resulting circuit, then no further processing is required for forming an interconnect layer. It is contemplated, however, that the unexposed regions **38** may not be sufficient in terms of insulation, dielectric value, structural integrity, durability, reliability, and/or the like to form a part of the final product. FIGS. **4a** through **4f** illustrate subsequent process steps that, if needed, remove unexposed portions **38** and replace them with dielectric material that may be more appropriate for the particular application. Although substrate **32** remains, for brevity and clarity substrate **32** is not illustrated in FIGS. **4a** through **4f**.

(36) Beginning with FIG. **4a**, the structure illustrated in FIG. **3f** is subjected to a process whereby plug **50** and remaining unexposed portions **38** are removed. The particular process(es) used to remove plug **50** and remaining unexposed portions **38** will depend upon the particular materials of these features. In a contemplated embodiment, etching, ashing, or the like can be employed to remove the features, resulting in opening **53**. Note that in the illustrated embodiment, the ashing/etching process(es) may result in also removing some or all of transparent cap layer **44**. This is a by-product of contemplated removal processes but it is not necessary that transparent cap layer **44** be removed. Conversely, it is illustrated that transparent cap layer **42** remains after the ashing/etching processes, but this too is simply a design choice and transparent cap layer **42** could likewise be removed (or not) during the ashing/etching process(es). Note also that opening **53** extends through ESL layers **34** and **36** in the illustrated embodiment—the result of the same or additional etching steps as used to remove plug **50** and/or remaining portions **38**. It is not necessary, however, that ESL layer **34** and **36** be removed in opening **53**, and in some embodiments (not illustrated) these layers remain at the bottom of opening **53**.

(37) Next, as illustrated in FIG. **4b**, a metal isolation liner **56** may optionally be formed to line sidewalls and the bottom of opening **53**. Metal isolation liner could be formed of a dielectric such as silicon oxide, silicon nitride, or the like, or in some embodiments metal isolation liner could be

formed of a metal barrier such as titanium, titanium nitride, tantalum, tantalum nitride, or the like. In embodiments wherein metal isolation liner **56** is a metal barrier, the liner is removed from the bottom of opening **53**, as shown in FIG. **4c**.

(38) Then, as shown in FIG. **4c**, lined opening **53** is filled or overfilled with a dielectric material **58**. The dielectric material and deposition technique is a matter of design choice, but it is contemplated that a flowable Extra Low K (ELK) dielectric material be employed because of the advantageous properties of good gap filling capability and low dielectric constant properties. If needed, excess dielectric material **58** can be removed through a CMP process, an etch back process, or other planarization process, resulting in formation of dielectric feature **60** electrically isolating conductive exposed regions **54**, as illustrated in FIG. **4d**. The result is a level of electrically conductive features formed of the exposed (now conductive) regions **54** that were formed from photo-sensitive layer **38**, electrically isolated and physically separated by dielectric features **60** formed in the gaps left behind when unexposed portions of photo-sensitive layer **38** were removed. While not necessary, optionally additional interconnect layers can be stacked atop the intermediate structure shown in FIG. **4d** by repeating the process steps illustrated in FIGS. **3a** through **4d**. For instance, as shown in FIG. **4e**, an optional capping layer **62** can be formed over conductive exposed portions **54** to protect them against subsequent processes and, as shown in FIG. **4f**, a next level of ESL layers **34'** and **36'** can be formed over capping layer **62** and dielectric feature **60**. In a contemplated embodiment, capping layer **62** could comprise cobalt, although other materials are similarly contemplated.

(39) Another embodiment process for converting a photo-sensitive layer into a layer with conductive regions is illustrated in FIGS. **5a** through **5j**. Beginning with FIG. **5a**, an intermediate state of manufacture is illustrated in which ESL layers **74** and **76** have been formed atop substrate **72**. Substrate **72** can be the same or similar to substrate **2** described above. In other words, substrate **72** can be a bulk semiconductor wafer, a multi-layer wafer such as an SOI substrate, could comprise silicon or other semiconductor materials, could be doped or undoped, etc. Further, as described above with reference to substrate **2**, substrate **72** is contemplated encompass various layers/features/structures as may be formed in and on the wafer including but not limited to transistors, capacitors, resistors, interconnects, insulating layers, dielectric layer, metallization layers, and the like.

(40) ESL layers **74** and **76** are deposited atop substrate **72**. These ESL layers could be the same or similar to ESL layers **34** and **36** discussed above, but such similarity is not a requirement for this embodiment. Additionally, a single ESL layer or more than two ESL layers could be employed, provided the ESL layer or layers offer sufficient protection to underlying layers/features/structures during subsequent processing steps as described in the following paragraphs. In this embodiment, ESL **74** is a layer of aluminum nitride, for instance, and ESL **76** is a layer of ODC, for example. Other materials are also within the contemplated scope of this disclosure, provided the materials provide sufficient protection for the underlying layers/features/structures of substrate **72** during subsequent processes as described in the following paragraphs.

(41) FIG. **5b** illustrates a photo-sensitive layer **78** is deposited on ESL layer **76**. In this embodiment, photo-sensitive layer **78** is a monomer such that its conductivity changes significantly upon sufficient exposure to light energy of certain wavelengths, and in this way the above teaching regarding photo-sensitive layer **38** is applicable to photo-sensitive layer **78**. For brevity and clarity, the above discussion regarding materials, properties, and processes for photo-sensitive layer **38** are incorporated herein with regard to photo-sensitive layer **78** by reference.

(42) In this embodiment, however, a photo-sensitive layer **78**, such as thiophene, is deposited, preferably using a spin-on deposition technique. Thiophene is but one example of an oligomer, i.e. a low molecular weight polymer.

(43) As illustrated in FIG. **5c**, cap layers are formed on photo-sensitive layer **78**. In this embodiment, only two transparent cap layers **82** and **84** are illustrated. Depending upon the

material selected, it may be possible to directly pattern transparent cap layers **82** and **84** without the use of additional capping layers. Alternatively, transparent cap layer **82** and **84** could be used in conjunction with additional cap layers such as cap layers **46** and **48** of FIG. **3c** in those applications where they are needed to more accurately pattern transparent cap layers **82** and **84**. These additional cap layers could be the same materials as described with reference to FIG. **3c**, or could be other materials provided they are compatible with transparent cap layers **82** and **84** and with the processes described in this embodiment.

(44) FIG. **5d** illustrates the results of patterning transparent cap layers **82** and **84** to form an opening therein, filling the opening with a plug **86**, and then exposing the device to a light energy source having a light energy schematically illustrated by arrows **88**. Assuming thiophene is the photo-sensitive layer **78**, light energy **88** should have a wavelength in the range of from about 200 nm to about 400 nm in order to activate the photo-sensitive properties of photo-sensitive layer **78**. Plug **86** should be opaque (not transparent) at these wavelengths in order to protect photosensitive material underlying the plug. Materials such as SiO₂, SiOC, and the like are transparent to light at these wavelengths and could be employed for plug **86**. Transparent cap layers **82** and **84** should be transparent or at least substantially transparent to these wavelengths to allow sufficient light energy to pass through them and expose (activate) the underlying regions of photo-sensitive layer **78**. Materials such as SiO₂, SiOC, and the like at a thickness range of from about 5 nm to about 70 nm are illustrative examples for transparent cap layers **82** and **84**.

(45) During the light exposure process, the region of photo-sensitive layer **78** covered by plug **86** is protected from (not exposed by) light energy **88**. By contrast, the regions of photo-sensitive layer **78** covered by transparent cap layers **82** and **84** are exposed to light energy **88** because transparent cap layers **82** and **84** are substantially transparent to the wavelengths of light energy **88**. In response to exposure to incoming light **88** in an illustrative range of from about 10 mJ/cm² to about 100 mJ/cm², the exposed portions of photo-sensitive layer **78** undergo a change in at least one material property and in this embodiment those exposed portions convert from dielectric to conductive regions **90** (for clarity, those portions of photo-sensitive layer **78** that remain unexposed will henceforth be called unexposed portions **78**, remaining portions **78**, or the like in contrast to those portions of photo-sensitive layer **78** that were exposed which will henceforth be called exposed regions **90**, conductive regions **90**, or the like).

(46) Exposed regions **90** (now conductive) can be in the form of contact plugs, pads, elongated wires, or other forms of interconnect and the remaining unexposed regions **78** can be in the form of a patterned insulating layer that electrically isolated the various exposed conductive regions **90** from one another. If the remaining unexposed regions **78** are sufficiently insulating to perform this function during operation of the resulting circuit, then no further processing is required for forming an interconnect layer. It is contemplated, however, that the unexposed regions **78** may not be sufficient in terms of insulation, dielectric value, structural integrity, durability, reliability, and/or the like to form a part of the final product. FIGS. **5e** through **5j** illustrate subsequent process steps that, if needed, remove unexposed portions **78** and replace them with dielectric material that may be more appropriate for the particular application.

(47) Beginning with FIG. **5e**, the structure illustrated in FIG. **5d** is subjected to a process whereby plug **86** and remaining unexposed portions **78** are removed. The particular process(es) used to remove plug **8** and remaining unexposed portions **78** will depend upon the particular materials of these features. In a contemplated embodiment, etching, ashing, or the like can be employed to remove the features, schematically illustrated by arrows **89** in FIG. **5e**. The result is opening **91**. Note that in the illustrated embodiment, the ashing/etching process(es) **89** may result in also removing some or all of transparent cap layer **84**. This is a by-product of contemplated removal processes but it is not necessary that transparent cap layer **84** be removed. Conversely, it is illustrated that transparent cap layer **82** remains after the ashing/etching processes, but this too is simply a design choice and transparent cap layer **82** could likewise be removed (or not) during the

ashing/etching process(es). Note also that opening **91** extends through ESL layers **74** and **76** in the illustrated embodiment—the result of the same or additional etching steps as used to remove plug **86** and/or remaining portions **78**. It is not necessary, however, that ESL layer **74** and **376** be removed in opening **91**, and in some embodiments (not illustrated) these layers remain at the bottom of opening **91**.

(48) Next, as illustrated in FIG. **5f**, a metal isolation liner **92** may optionally be formed to line sidewalls and the bottom of opening **91**. Metal isolation liner could be formed of a dielectric such as silicon oxide, silicon nitride, or the like, or in some embodiments metal isolation liner could be formed of a metal barrier such as titanium, titanium nitride, tantalum, tantalum nitride, or the like. In the illustrated embodiment, FIG. **5g**, metal isolation liner **92** is removed from the bottom portion of opening **91**. In some embodiments, however, metal isolation liner **92** could be left remaining at the bottom of opening **91**.

(49) Then, as shown in FIG. **5g**, lined opening **91** is filled or overfilled with a dielectric material **96**. The dielectric material and deposition technique is a matter of design choice, but it is contemplated that a flowable Extra Low K (ELK) dielectric material be employed because of the advantageous properties of good gap filling capability and low dielectric constant properties. If needed, excess dielectric material **96** can be removed through a CMP process, an etch back process, or other planarization process, resulting in formation of dielectric feature **97** electrically isolating conductive exposed regions **90**, as illustrated in FIG. **5g**. The result is a level of electrically conductive features formed of the exposed (now conductive) regions **90** that were formed from photo-sensitive layer **78**, electrically isolated and physically separated by dielectric features **97** formed in the gaps left behind when unexposed portions of photo-sensitive layer **78** were removed. While not necessary, optionally additional interconnect layers can be stacked atop the intermediate structure shown in FIGS. **5i** and **5j** by repeating the process steps illustrated in FIGS. **5a** through **5d**. For instance, as shown in FIG. **5i**, an optional capping layer **98** can be formed over conductive exposed portions **90** to protect them against subsequent processes and, as shown in FIG. **5i**, and a next level of ESL layers **74'** and **76'** can be formed over capping layer **98** and dielectric feature **97**. In a contemplated embodiment, capping layer **98** could comprise cobalt, although other materials are similarly contemplated.

(50) FIGS. **6** and **7** provide additional information about the chemical composition of the photo-sensitive layers used in the embodiments described herein.

(51) One general aspect of embodiments disclosed herein includes forming a photo-sensitive material over a substrate. The method also includes forming a cap layer over the photo-sensitive material. The method also includes patterning the cap layer. The method also includes using the patterned cap layer, selectively exposing a first portion of the photo-sensitive material to a pre-selected light wavelength to change at least one material property of the first portion of the photo-sensitive material, while preventing a second portion of the photo-sensitive material from being exposed to the pre-selected light wavelength. The method also includes and performing one, but not both of the following steps: removing the first portion of the photo-sensitive material and forming in its place a conductive element at least partially surrounded by the second portion of the photo-sensitive material. The method also includes removing the second portion of the photo-sensitive material and forming from the first portion of the photo-sensitive material a conductive element electrically connecting two or more portions of a circuit.

(52) Another general aspect of embodiments disclosed herein includes depositing on a substrate a photo-sensitive material. The method also includes selectively exposing first portions of the photo-sensitive material to a pre-selected wavelength of light in order to change at least one material property of the first portions of the photo-sensitive material, while not exposing second portions of the photo-sensitive material to the pre-selected wavelength of light. The method also includes incorporating the photo-sensitive material into an electrical interconnection of an integrated circuit.

(53) Yet another general aspect of embodiments disclosed herein includes a substrate. The device

also includes an electrical interconnection circuit formed over the substrate, the electrical interconnection circuit having a least two levels, each of the two levels including: a layer of photo-sensitive material, the layer having regions of conductive photo-sensitive material, and the layer further having trenches therein. The device also includes and an insulating material formed within the trenches and electrically insulating the regions of conductive photo-sensitive material.

(54) One or more embodiments disclosed herein may include a method comprising providing a substrate, depositing a first etch stop layer over the substrate, and depositing a second etch stop layer over the first etch stop layer, and blanket depositing a photo-sensitive layer over the second etch stop layer. The method may further include forming a plurality of cap layers over the photo-sensitive layer, patterning the plurality of cap layers to create an opening in the plurality of cap layers, and exposing a first part of the photo-sensitive layer to light of a pre-selected wavelength to change a material property of the first part of the photo-sensitive layer, while masking a second part of the photo-sensitive layer from exposure to the light of the pre-selected wavelength. The method may further include steps of either (a) removing the first part of the photo-sensitive layer and forming a conductor in its place, or (b) removing the second part of the photo-sensitive layer and forming an insulator in its place.

(55) Other embodiments disclosed herein may include a method of providing a substrate, forming a first and a second etch stop layer over the substrate, depositing a photo-sensitive layer, over the second etch stop layer, and blanket depositing cap layers over the photo-sensitive layer, the cap layers being transparent to a pre-selected wavelength of light. Additionally, the method may include patterning the cap layers to form an opening therein, filling the opening with a plug, exposing the photo-sensitive layer to light energy of the pre-selected wavelength through the cap layers to convert exposed portions of the photo-sensitive layer to conductive regions, and removing unexposed portions of the photo-sensitive layer to form trenches, and filling the trenches with a material.

(56) Other embodiments disclosed herein may include a method comprising: depositing a photo-sensitive layer over a substrate, the substrate including an etch stop layer formed thereon, and forming a patterned mask layer over the photo-sensitive layer, the patterned mask layer masking portions of the photo-sensitive layer while leaving second portions of the photo-sensitive layer un-masked. Steps of this method may also include exposing the un-masked portions of the photo-sensitive layer to a light of a pre-selected wavelength, wherein the exposure changes an electrical conductivity characteristic of the un-masked portions, or a resistance to a pre-selected etch process of the un-masked portions. When the exposure changes the electrical conductivity of the un-masked portions, the method includes forming an insulator electrically isolating adjacent un-masked portions. When the exposure changes the resistance to the pre-selected etch process of the un-masked portions, the method includes etching the un-masked portions to form an opening, and filling the opening with a conductive material.

(57) Some embodiments disclosed herein provide for a method of forming a device, which includes forming over a substrate a layer of photo-sensitive material, the photo-sensitive material having a first dielectric value. The method also includes forming a mask over a first region of the photo-sensitive material while leaving a second region of the photo-sensitive material exposed. The method also includes changing the dielectric value of the second region of the photo-sensitive material to a second, lower dielectric value by exposing the second region of the photo-sensitive material to a pre-selected wavelength of light energy, while leaving the dielectric value of the first region of the photo-sensitive material unchanged.

(58) Other embodiments disclosed herein provide for method including forming a photo-sensitive material over a substrate, and forming a patterned cap layer over the photo-sensitive material. The method also includes using the patterned cap layer, selectively exposing a first portion of the photo-sensitive material to a pre-selected wavelength of light energy to change at least one material property of the first portion of the photo-sensitive material, while leaving the at least one material

property of a second portion of the photo-sensitive material unchanged by the exposure process. The method also includes preventing a second portion of the photo-sensitive material from being exposed to the pre-selected wavelength of light energy.

(59) Yet other embodiments disclosed herein provide for a method of forming a device, which method includes forming over a substrate a layer of photo-sensitive material and forming over a first region of the photo-sensitive material a cap layer, the cap layer being transparent to a pre-selected wavelength of light energy. The method also includes forming over a second region of the photo-sensitive material a plug, the plug being opaque to the pre-selected wavelength of light energy. The method also includes converting the first region of the photo-sensitive material from a dielectric to a conductor by exposing the first region to the pre-selected wavelength of light energy, while leaving the second region of the photo-sensitive material unconverted as a dielectric.

(60) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: providing a substrate; depositing a first etch stop layer over the substrate; depositing a second etch stop layer over the first etch stop layer; blanket depositing a photo-sensitive layer over the second etch stop layer; forming a plurality of cap layers over the photo-sensitive layer; patterning the plurality of cap layers to create an opening in the plurality of cap layers; exposing a first part of the photo-sensitive layer to light of a pre-selected wavelength to change a material property of the first part of the photo-sensitive layer, while masking a second part of the photo-sensitive layer from exposure to the light of the pre-selected wavelength; and either (a) removing the first part of the photo-sensitive layer and forming a conductor in its place, or (b) removing the second part of the photo-sensitive layer and forming an insulator in its place.
2. The method of claim 1, wherein: the step of blanket depositing the photo-sensitive layer includes depositing a layer of material selected from the group consisting of thiophene, phenyl, thiophenyl, an I-R material, and a P-R material, where I is Iodonium salt, P is Phosphorous, and R represents C_xH_y or a $COOH$ organic group; the step of exposing the first part of the photo-sensitive layer to the light of the pre-selected wavelength includes exposing the first part of the photo-sensitive layer to an incoming light with a wavelength of from 200 nm to 400 nm; the material property of the first part of the photo-sensitive layer that is changed by exposure to the incoming light of the pre-selected wavelength is resistance to a pre-selected etch process of the first part of the photo-sensitive layer; and the method further including subjecting the first part of the photo-sensitive layer to the pre-selected etch process to remove the first part of the photo-sensitive layer.
3. The method of claim 2, wherein the resistance of the first part of the photo-sensitive layer to the pre-selected etch process is decreased by a factor in a range of from $2\times$ to $50\times$, as a result of the step of exposing the first part of the photo-sensitive layer to the incoming light with a wavelength of from 200 nm to 400 nm of a pre-selected wavelength.
4. The method of claim 2, wherein the step of subjecting the first part of the photo-sensitive layer to the pre-selected etch process forms a second opening, and further comprising filling the second opening with a conductive material.
5. The method of claim 1, further comprising: filling the opening in the plurality of cap layers with a plug, the plug being transparent to light of the pre-selected wavelength, and further comprising:

wherein the step of forming a plurality of cap layers over the photo-sensitive layer includes depositing at least one cap layer that is transparent to light of the pre-selected wavelength; wherein the step of blanket depositing the photo-sensitive layer includes depositing a layer of material selected from the group consisting of thiophene, phenyl, thiophenyl, an I-R material, and a P-R material, where I represents Iodonium salt, P represents Phosphorous, and R represent C_xH_y or a COOH organic group; wherein the step of exposing the first part of the photo-sensitive layer to light of a preselected wavelength includes exposing the first part of the photo-sensitive layer to an incoming light with a wavelength in the Extreme Ultra Violet (EUV) range; wherein the material property of the first part of the photo-sensitive layer that is changed by exposure to the incoming light of the pre-selected wavelength is electrical conductivity; and after the step of exposing the first part of the photo-sensitive layer to light of the pre-selected wavelength, removing the plug and the second part of the photo-sensitive layer to form a second hole; and filling the second hole with an insulator feature.

6. The method of claim 5, wherein the step of exposing the first part of the photo-sensitive layer to light of the pre-selected wavelength converts the first part of the photo-sensitive layer from a dielectric to a conductor.

7. The method of claim 5, wherein the step of filling the second hole with the insulator feature includes lining the second hole with a barrier layer.

8. The method of claim 5, wherein the step of blanket depositing the photo-sensitive layer includes depositing low molecular weight polymer using a spin-on deposition process before the step of filling the hole with the insulator features.

9. A method comprising: providing a substrate; forming a first and a second etch stop layer over the substrate; depositing a photo-sensitive layer over the second etch stop layer; blanket depositing cap layers over the photo-sensitive layer, the cap layers being transparent to a pre-selected wavelength of light; patterning the cap layers to form an opening therein; filling the opening with a plug; exposing the photo-sensitive layer to light energy of the pre-selected wavelength through the cap layers to convert exposed portions of the photo-sensitive layer to conductive regions; and removing unexposed portions of the photo-sensitive layer to form trenches; and filling the trenches with a material.

10. The method of claim 9, wherein the step of depositing a photo-sensitive layer includes depositing a layer of material selected from the group consisting of thiophene, phenyl, thiophenyl, an I-R material, and a P-R material, where I represents Iodonium salt, P represents Phosphorous, and R represents C_xH_y or a COOH organic group; and wherein the pre-selected wavelength is in the deep Extreme Ultra Violet (EUV) range.

11. The method of claim 9, further comprising performing a plasma pretreatment before the step of depositing a photo-sensitive layer over the second etch stop layer.

12. The method of claim 11, wherein the plasma pretreatment is an $O_{2.0}$ or a $CO_{x.0}$ plasma process.

13. The method of claim 9, wherein at least one of the cap layers is a material selected from the group consisting of $SiO_{2.0}$ and $SiOC$.

14. The method of claim 9, wherein the step of filling the opening with a plug comprises selectively depositing on surfaces of the photo-sensitive layer exposed by the opening in the cap layers.

15. The method of claim 9, further comprising forming a third etch stop layer over the conductive regions and the material filling the trench.

16. A method comprising: depositing a photo-sensitive layer over a substrate, the substrate including an etch stop layer formed thereon; forming a patterned mask layer over the photo-sensitive layer, the patterned mask layer masking portions of the photo-sensitive layer while leaving second portions of the photo-sensitive layer un-masked; exposing the un-masked portions of the photo-sensitive layer to a light of a pre-selected wavelength, wherein the exposure changes

an electrical conductivity characteristic of the un-masked portions, or a resistance to a pre-selected etch process of the un-masked portions; when the exposure changes the electrical conductivity of the un-masked portions, forming an insulator electrically isolating adjacent un-masked portions; and when the exposure changes the resistance to the pre-selected etch process of the un-masked portions, etching the un-masked portions to form an opening, and filling the opening with a conductive material.

17. The method of claim 16, wherein the step of forming a patterned mask layer over the photo-sensitive layer includes: blanket depositing over the photo-sensitive layer a masking layer that is transparent to the pre-selected wavelength; patterning the masking layer to form therein an opening exposing a portion of the photo-sensitive layer; and forming in the opening a plug that blocks light of the pre-selected wavelength.

18. The method of claim 17, further comprising exposing the un-masked portions of the photo-sensitive layer to a light of a pre-selected wavelength through the masking layer.

19. The method of claim 16, wherein the step of forming a patterned mask layer over the photo-sensitive layer includes: blanket depositing over the photo-sensitive layer a masking layer that blocks light of the pre-selected wavelength; and patterning the masking layer to form therein an opening exposing a portion of the photo-sensitive layer.

20. The method of claim 19, wherein the step of forming the insulator electrically isolating adjacent un-masked portions, includes using the un-masked portions of the photo-sensitive layer as an insulator.
